THE DIRECT IMAGE ON PRESHEAVES IS RIGHT ADJOINT TO THE INVERSE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's argument and in fact offers two complete routes: the first reproduces the companion argument and the second carries out the translation to the opposite category and applies the known adjunction. The sentences pointing to the companion proof are accompanied by the explicit identifications, and both are preserved verbatim inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The adjunction for the other pair of functors on presheaves. Same chapter, Lemma lemma-adjoints-u. Supplies the argument which is reproduced, and the dual computation.
- The unit of the other adjunction. Same chapter, Lemma lemma-recover-pu. Identifies the map used in the construction.
- Presheaves with values in the opposite category. Same chapter, Remark remark-functoriality-presheaves-values. Supplies the adjunction for presheaves with values in the opposite category used in the alternative argument.

Source reference. Original lemma 00VE: sources/stacks-sites/pinned-original-source.tex, source lines 4373–4373. Navigation pointer invoked at proof line 4373; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XG: sources/stacks-sites/pinned-original-source.tex, source lines 4375–4375. Navigation pointer invoked at proof line 4375; the counted span contains the pointer only and the cited result is not reproduced inside it, so

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Source reference. Original remark 00VF: sources/stacks-sites/pinned-original-source.tex, source lines 4385–4385. Navigation pointer invoked at proof line 4385; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *The functor ${}_pu$ is a right adjoint to the functor u^p. In other words the formula*

$$\mathrm{Mor}_{PSh(\mathcal{C})}(u^p\mathcal{G}, \mathcal{F}) = \mathrm{Mor}_{PSh(\mathcal{D})}(\mathcal{G}, {}_pu\mathcal{F})$$

holds bifunctorially in $\mathcal{F}$ and $\mathcal{G}$.

Proof. This is proved in exactly the same way as the proof of Lemma 00VE. We note that the map $u^p{}_pu\mathcal{F} \to \mathcal{F}$ from Lemma 00XG is the map that is used to go from the right to the left.

Alternately, think of a presheaf of sets $\mathcal{F}$ on $\mathcal{C}$ as a presheaf $\mathcal{F}'$ on $\mathcal{C}^{opp}$ with values in $Sets^{opp}$, and similarly on $\mathcal{D}$. Check that $({}_pu\mathcal{F})' = u_p(\mathcal{F}')$, and that $(u^p\mathcal{G})' = u^p(\mathcal{G}')$. By Remark 00VF we have the adjointness of u_p and u^p for presheaves with values in $Sets^{opp}$. The result then follows formally from this. □